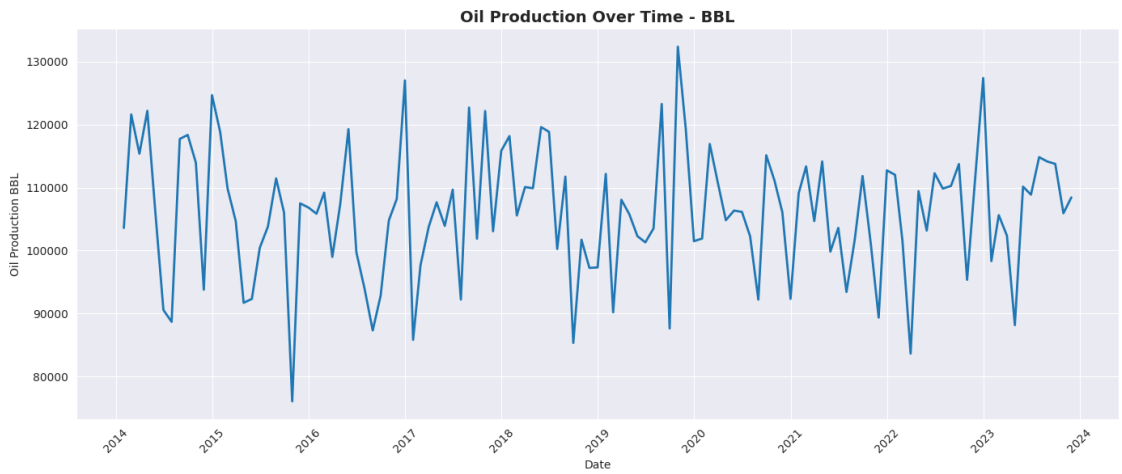
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import t
from sklearn.linear_model import Li
from sklearn.ensemble import Random
from sklearn.metrics import r2_scor

# 1. الكولاب – 10 سنين بيانات شهرية
np.random.seed(42)
dates = pd.date_range(start='2014-0

df = pd.DataFrame({
    'Date': dates,
    'Oil_Price_USD': np.random.unif
    'Rigs_Count': np.random.randint
    'Temperature_C': np.random.unif
    'Maintenance_Days': np.random.r
})

# نسوي علاقة واقعية للانتاج
df['Oil_Production_BBL'] = (
    50000 +
    df['Oil_Price_USD'] * 300 +
    df['Rigs_Count'] * 80 -
    df['Maintenance_Days'] * 2000 +
    np.random.normal(0, 5000, len(d
).astype(int)

print("تم توليد الداتا ✅")
print("شكل الداتا:", df.shape)
print("\nأسماء الأعمدة:", df.columns
print("\nأول 5 صفوف:")
print(df.head())

# 2. تحضير الداتا للموديل
X = df[['Oil_Price_USD', 'Rigs_Coun
y = df['Oil_Production_BBL']
X_train, X_test, y_train, y_test =

# 3. نجرب موديلين ونقارن
models = {
    "Linear Regression": LinearRegr
    "Random Forest": RandomForestRe
}
```

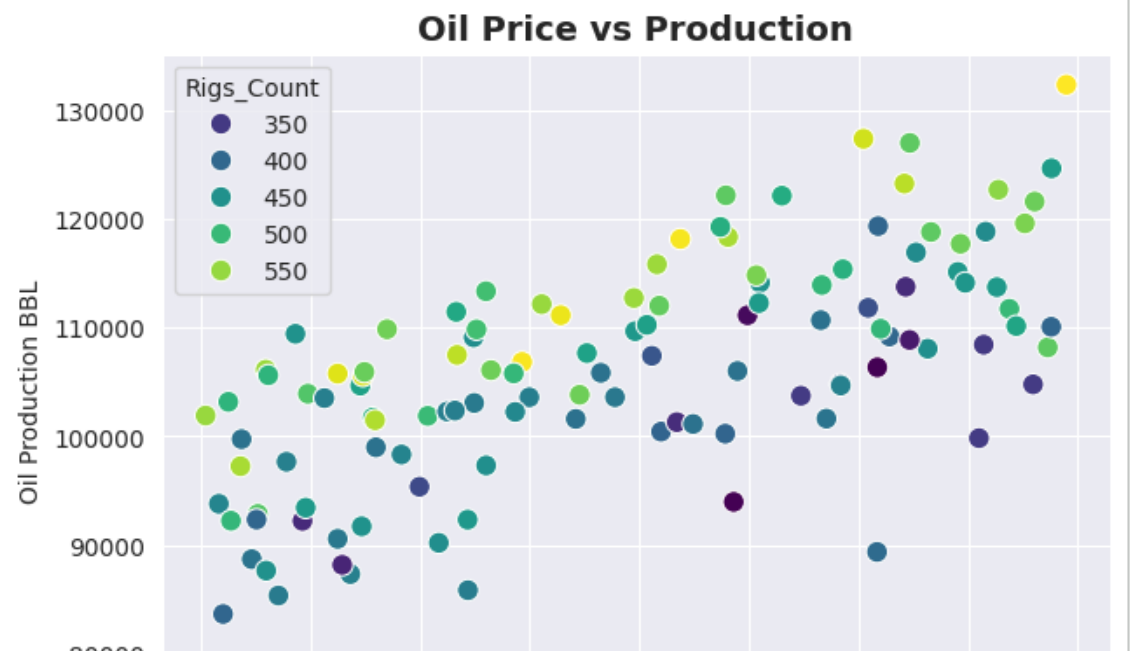
```
for name, model in models.items():
    model.fit(X_train, y_train)
    preds = model.predict(X_test)
    r2 = r2_score(y_test, preds)
    mae = mean_absolute_error
```

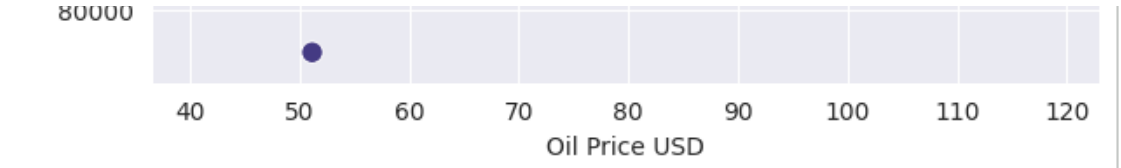
```
/tmp/ipykernel_467/730634818.py:11: FutureWarning: 'M' is depre
  dates = pd.date_range(start='2014-01-01', end='2023-12-01', f
تم توليد الداتا ✅
شكل الداتا: (119, 6)

أسماء الأعمدة: ['Date', 'Oil_Price_USD', 'Rigs_Count', 'Temperat

أول 5 صفوف:
        Date  Oil_Price_USD  Rigs_Count  Temperature_C  Mainter
0 2014-01-31      69.963210         427      41.761397
1 2014-02-28     116.057145         530      33.934159
2 2014-03-31      98.559515         489      38.844339
3 2014-04-30      87.892679         524      30.079113
4 2014-05-31      52.481491         582      32.307117

   Oil_Production_BBL
0              103599
1              121630
2              115395
3              122205
4              105777
```

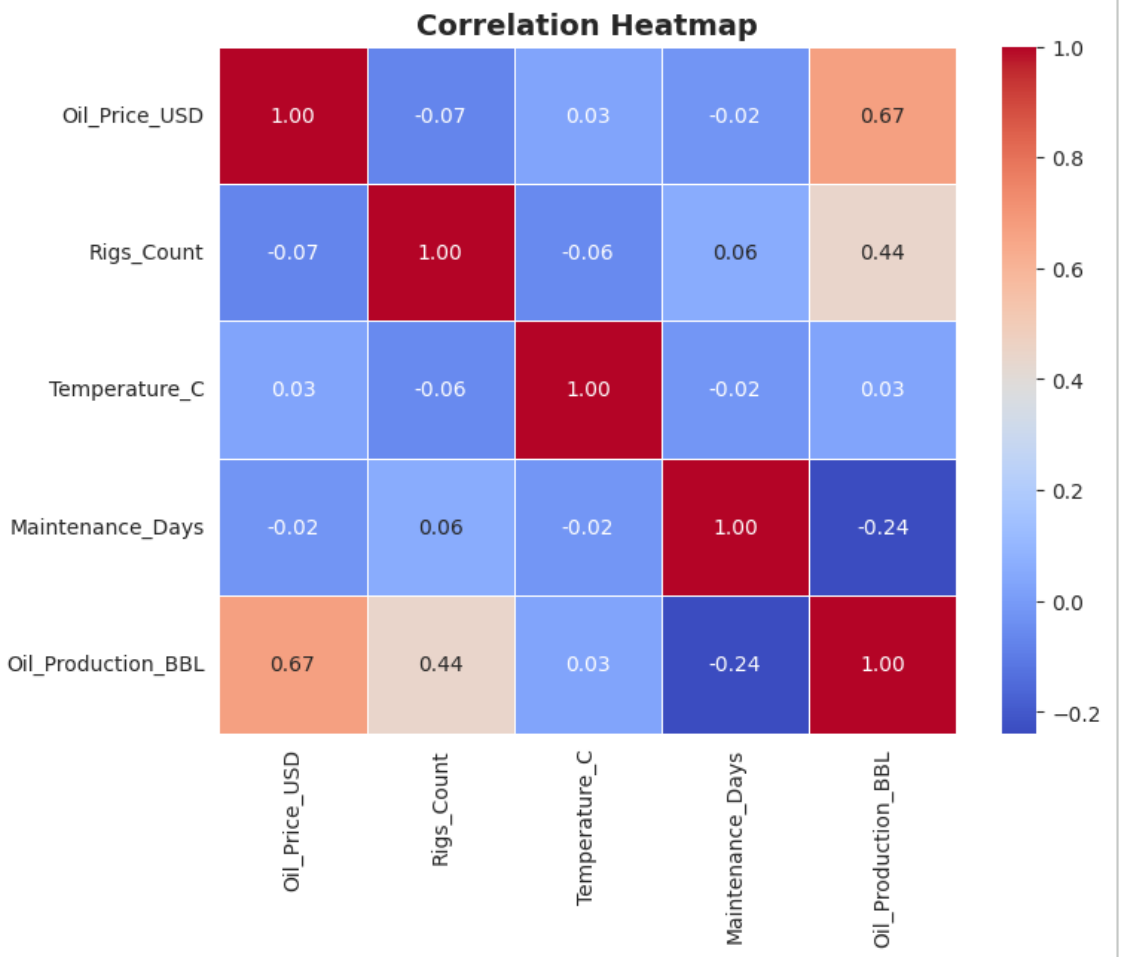
```
import matplotlib.pyplot as plt
import seaborn as sns

# نخلي الشكل احترافي
sns.set_style("darkgrid")
plt.rcParams['figure.figsize'] = (12, 5)

# 1. رسم خط زمني للإنتاج
plt.figure(figsize=(14,6))
plt.plot(df['Date'], df['Oil_Production_BBL'], color='#1f77b4'
plt.title('Oil Production Over Time - BBL', fontsize=14, fontw
plt.xlabel('Date')
plt.ylabel('Oil Production BBL')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

# 2. علاقة سعر النفط بالإنتاج
plt.figure(figsize=(7,5))
sns.scatterplot(x='Oil_Price_USD', y='Oil_Production_BBL', dat
plt.title('Oil Price vs Production', fontsize=14, fontweight='
```

```
plt.xlabel('Oil Price USD')
plt.ylabel('Oil Production BBL')
plt.show()

# 3. Heatmap للارتباط بين الأعمدة
plt.figure(figsize=(8,6))
corr = df.select_dtypes(include=np.number).corr()
sns.heatmap(corr, annot=True, cmap='coolwarm', fmt=".2f", line
plt.title('Correlation Heatmap', fontsize=14, fontweight='bold
plt.show()

# 4. تأثير أيام الصيانة على الإنتاج
plt.figure(figsize=(7,5))
sns.boxplot(x='Maintenance_Days', y='Oil_Production_BBL', data
plt.title('Maintenance Days Impact on Production', fontsize=14
plt.xlabel('Maintenance Days')
plt.ylabel('Oil Production BBL')
plt.show()
```

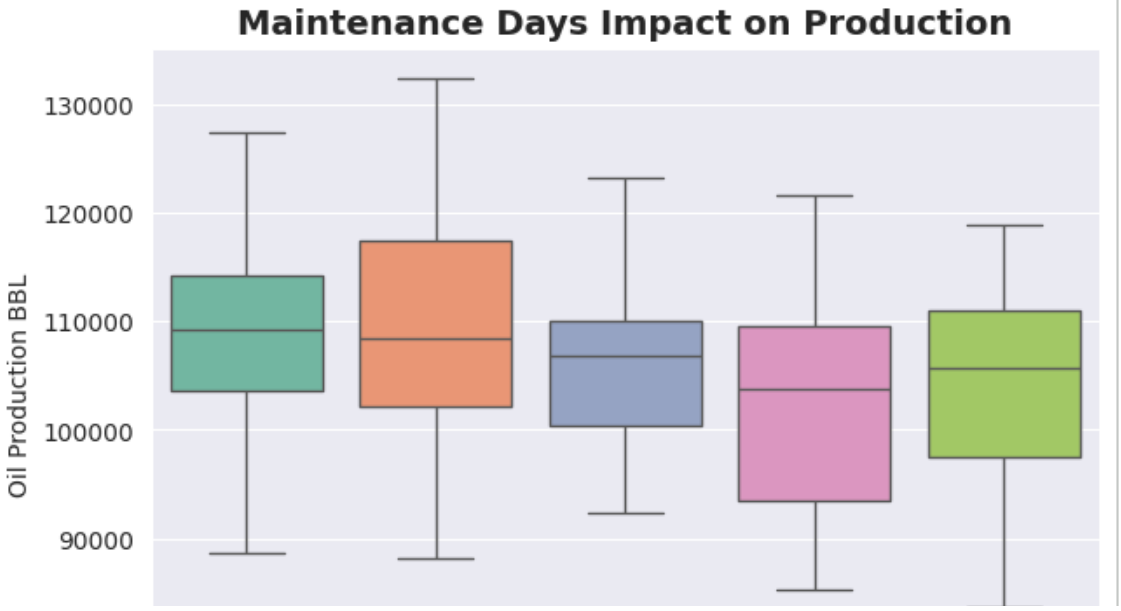








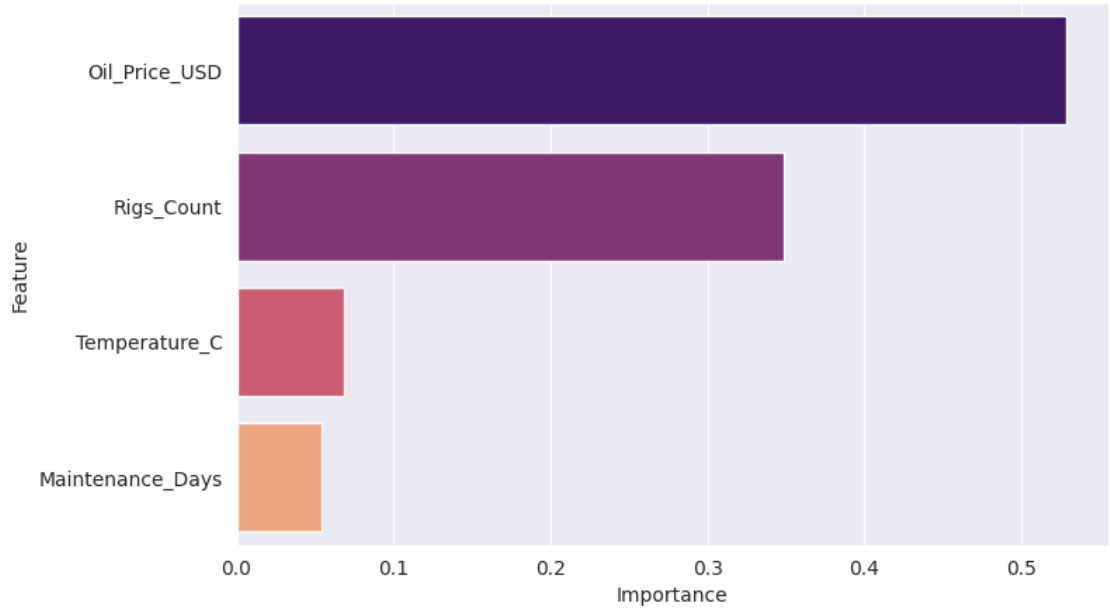

```
/tmp/ipykernel_467/1456204514.py:35: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and wil

  sns.boxplot(x='Maintenance_Days', y='Oil_Production_BBL', dat
```




```
from sklearn.ensemble import RandomForestRegressor
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd

# 1. ندرب Random Forest من جديد عشان نتأكد انه موجود
rf_model = RandomForestRegressor(n_estimators=100, random_stat
rf_model.fit(X_train, y_train)

# 2. نجيب أهمية كل ميزة
importances = rf_model.feature_importances_
features = X.columns

feat_imp = pd.DataFrame({
    'Feature': features,
    'Importance': importances
}).sort_values('Importance', ascending=False)
```

```
print("أهمية المتغيرات:")
print(feat_imp)

# 3. نرسمها
plt.figure(figsize=(8,5))
sns.barplot(x='Importance', y='Feature', data=feat_imp, hue='F
```

```
أهمية المتغيرات:
            Feature  Importance
0     Oil_Price_USD    0.528700
1        Rigs_Count    0.349012
2     Temperature_C    0.068341
3  Maintenance_Days    0.053946
<Axes: xlabel='Importance', ylabel='Feature'>
```




```
# Cell لحفظ كل الصور مرة وحدة
import matplotlib.pyplot as plt
import seaborn as sns

# 0. لازم نعرف المصفوفة بالأول
corr_matrix = df.corr() # <-- السطر

# 1. Feature Importance
plt.figure(figsize=(8,5))
```

```
plt.figure(figsize=(8,5))
sns.barplot(x='Importance', y='Feat
plt.title('Feature Importance for C
plt.tight_layout()
plt.savefig('feature_importance.png
plt.close()

# 2. Correlation Heatmap
plt.figure(figsize=(8,6))
sns.heatmap(corr_matrix, annot=True
plt.title('Correlation Heatmap')
plt.tight_layout()
plt.savefig('correlation_heatmap.pn
plt.close()

# 3. Price vs Production
plt.figure(figsize=(8,5))
sns.scatterplot(x='Oil_Price_USD',
plt.title('Oil Price vs Production'
plt.tight_layout()
plt.savefig('price_vs_production.pn
plt.close()

# 4. Rigs vs Production
plt.figure(figsize=(8,5))
sns.scatterplot(x='Rigs_Count', y='
plt.title('Rigs Count vs Production
plt.tight_layout()
plt.savefig('rigs_vs_production.png
plt.close()

# 5. Maintenance Boxplot
plt.figure(figsize=(8,5))
sns.boxplot(x='Maintenance_Days', y
plt.title('Maintenance Days vs Prod
plt.tight_layout()
plt.savefig('maintenance_boxplot.pn
plt.close()

print("⌨ للملفات على اليسار! جاح
```

تم حفظ كل الصور بنجاح! روحي للملفات على اليسار

```
import matplotlib.pyplot as plt
import seaborn as sns
from google.colab import files

# 1. نعيد تعريف feat_imp بالأول
feat_imp = pd.Series(rf_model.feature_importances_, index=X.co
feat_imp = feat_imp.reset_index()
feat_imp.columns = ['Feature', 'Importance']

# 2. نحفظ Feature Importance – أهم صورة
plt.figure(figsize=(8,5))
sns.barplot(x='Importance', y='Feature', data=feat_imp, hue='F
plt.title('Feature Importance for Oil Production', fontsize=14
plt.tight_layout()
plt.savefig('feature_importance.png', dpi=300, bbox_inches='ti
plt.close()

# 3. نحفظ Correlation Heatmap
plt.figure(figsize=(8,6))
sns.heatmap(df.corr(), annot=True, cmap='coolwarm', fmt='.2f')
plt.title('Correlation Heatmap')
plt.tight_layout()
plt.savefig('correlation_heatmap.png', dpi=300, bbox_inches='t
plt.close()

# 4. تحميل تلقائي
files.download('feature_importance.png')
files.download('correlation_heatmap.png')

print("تم! بينزلوا صورتين الحين: feature_importance.png و corr
```

```
تم! بينزلوا صورتين الحين: feature_importance.png و correlation_
```